GROUP NUMBER: 1 .
GROUP NAME: .
GROUP MEMBERS:

Name	Group Role(s)
Dan Caverly	Project Lead
Matthew Drummond	Data Architect
Michael Hicks	Data Scientist
Sean Stack	Communication Liaison

Credit Card Fraud Detection Dataset
A fully functional Git repository has been established for the project. URL: https://github.com/Lazyboy2002/EAS-345-Credit-Card-Fraud-Model/tree/main **Maintenance and Access:** The repository is jointly maintained by the Project Lead (Dan Caverly) and the Data Architect / Scribe (Matthew Drummond). All team members have full access and are able to clone, pull, commit, and push changes using Git. Each member has demonstrated the ability to use Git commands and navigate the repository. **Branch Strategy:** At this stage of the project, all development is taking place on the main branch. No additional branching strategies are in use. **Repository Issues:** There are no current issues or errors with the repository. It is functioning as intended and is being used for version control and documentation of the full modeling process.
Source Organization: The dataset originates from the IEEE Computational Intelligence Society (IEEE-CIS) **URL/Download Instructions:** Original data is provided through the IEEE-CIS Fraud Detection Kaggle competition, which can be found at https://www.kaggle.com/competitions/ieee-fraud-detection/data. Because of Kaggle's API restrictions, data must be downloaded via the Kaggle Command Line Interface (CLI). A Kaggle account is required.

Dataset Description:
The competition includes five files:

- train_transaction.csv
- train_identity.csv
- test_transaction.csv
- test_identity.csv
- sample_submission.csv

For this project, only train_transaction.csv and train_identity.csv were used. The test sets (test_transaction.csv and test_identity.csv) were excluded because they contain no target labels, and the competition is no longer active. The truth behind the test datasets is unavailable.
train_transaction.csv and train_identity.csv were converted to RDS file format so that they could be stored on the repository as convenience. Preprocessing can take place without any external downloads, as the raw csv files were too large to store on GitHub.

Dataset Characteristics:
train_transaction.csv contains 394 features (categorical, numeric, anonymized) and 590,541 transactions. It has a binary target variable isFraud. train_identity.csv contains 41 features and 144,234 entries. These two datasets were merged using a left join on TransactionID, resulting in an integrated dataset with 434 features.

Special Instructions:
Beyond requiring a Kaggle account and the Kaggle CLI, there are no additional special requirements for accessing or using the data.

Modifications to Original Data:
The team performed a multi-step preprocessing pipeline:

1. Data Integration: Left join of transaction and identity datasets; removal of features with >95% missingness
2. Outlier and Zero-Transaction Handling
3. Missing Value Treatment
4. Categorical Feature Encoding
5. Feature Engineering
6. Feature Selection
7. Preprocessed Dataset Output

After preprocessing, the working dataset contained 331 features.
Preprocessed RDS files (train_set.rds, test_set.rds, val_set.rds) are stored in the repository at data/.

Repository Contents:
All source code required for data processing, model training, and model testing is included in the repository:

- src/01_preprocess.R
- src/02_train_model.R
- src/03_test_model.R
- output/xgb_fraud.model (trained XGBoost model)

Data Availability:
All datasets needed to run the model are included as RDS files in the Git repository:

- data/train_set.rds
- data/val_set.rds
- data/test_set.rds

No external downloads are required; all files are version-controlled and included.
Team Ability to Run the Model
All team members have successfully run the preprocessing, model training, and model testing scripts. Each member confirmed that they can reproduce model outputs on their own machine.

Documentation:
A complete README.md is present in the repository. It includes:

- Required R packages
- Environment setup
- Instructions for running scripts using R Studio
- Directory structure
- Order of execution of scripts

The documentation is sufficient for any new contributor to run the project end-to-end.
Reproducibility
The modeling pipeline has been executed on multiple machines, and results are reproducible and consistent across environments.